IIT Pokerbots

Sneak Peek Hold'em

Bot Architecture & Strategy

IIT Pokerbots Competition 2026 · Final Presentation

Team Poker Noobs · 10th Place · Qualified for Finals

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Agenda

Problem: Sneak Peek Hold'em

Standard No-Limit Texas Hold'em + one asymmetric information auction after the flop

Game Rules

- 1,000 rounds × 5,000 chips/round
- Small blind: 10 | Big blind: 20
- Pre-flop betting (standard)
- Flop dealt → AUCTION phase
- Post-flop betting continues
- Turn → betting → River → betting
- Best 5-card hand wins the pot

The Auction

Sealed-bid	Both players bid simultaneously (no peeking)
Vickrey	Winner pays the LOWER bid into the pot
Reward	Winner sees ONE random opponent hole card
Tie rule	Both pay their bid; both see one opp card
Loser	Receives zero information — plays blind

System Architecture

Four tightly coupled modules — the bot decides every action in under 20 ms

Feature Extractor

28 hand-crafted poker features

MLP Equity Model

Neural net trained on 200k samples

Betting Engine

Game-theoretic raise/call/fold

Auction Bidder

Vickrey + trap logic

Opponent Model (β)

Bayesian EMA tracker — estimates bluff rate β,
mean bid μ, and aggression level in real time

Action Output

Fold | Check | Call | Raise(size) | Bid(amount)

MLP Equity Model

28-feature NN trained on 200k Monte Carlo equity samples across all 5 game states

Network Architecture

Input:	28 features
Hidden 1:	256 units + ReLU
Hidden 2:	128 units + ReLU
Hidden 3:	64 units + ReLU
Output:	1 × Sigmoid → P(win)

Training Setup

Samples:	200,000 (5 state types)
MC iters:	5,000 rollouts per label
Optimizer:	Adam · LR = 1e-3
Loss:	Mean Squared Error (MSE)
Epochs:	500 with early stopping
Device:	GPU (Google Colab T4)

28 Input Features

Hole Cards (5)
Rank values, suitedness, pairing, gap size

Board (7)
Hi/lo/avg rank, board pairing, flush/straight density

Hand Strength (5)
Made flush/straight, draw proximity, max kind

Relative Strength (4)
Top pair, overpair, board dominance

Opponent Info (5)
Revealed card rank, flush zone, board pairing

Eval7 tier (1)
Ground-truth 8-class hand category (final signal)

Board state (1)
Street indicator — how many community cards dealt

Equity Model — Accuracy & Benchmarks

Pre-flop validated against 100,000-iteration Monte Carlo ground truth

~1.8%
Pre-flop MAE
vs 100k MC

~82%
< 2% Error
of samples

~97%
< 5% Error
of samples

5
State Types
game stages

Accuracy by Game State

State	n_hole	n_board	n_opp	MAE
Pre-flop	2	0	0	~1.8%
Flop	2	3	0	~2.1%
Turn	2	4	0	~1.9%
Flop + Info	2	3	1	~2.4%
Turn + Info	2	4	1	~2.2%

Key MLP Properties

- Embedded in bot as base64-compressed weights (no file I/O)
- Runs in < 1 ms per inference — fast enough for 20s total budget
- Adjusted by σ_board (volatility) for risk awareness
- River accuracy was weaker — known limitation (see Collapse slide)

Mathematical Betting Framework

Every raise, call, and fold decision flows from a unified analytical formula

$$W = E_{adj} \times (1 - \beta) + 1.0 \times \beta \qquad \text{(True Win Probability)}$$

$$X = \max(0, 2W - 1) \qquad \text{(Advantage score)}$$

$$R_{cap} = S_{eff} \times I_{mod} \times X^{3.5}/(1.05-X) \times (1 - 0.5\beta) \qquad \text{(Raise cap)}$$

$$C_{cap} = R_{cap} + \beta \times \mu_{bluff} \times 1.5 \qquad \text{(Call cap)}$$

$$x^* = P \times [(1-W)/(2W-1) + \beta] \qquad \text{(Opening raise)}$$

W — True Win Probability

Bluff-weighted equity. If opponent bluffs (β↑), our true win rate is higher than the card equity alone suggests.

X — Advantage Score

Transforms W into a [0,1] advantage. At W=0.5 (coin flip), X=0. At W=1.0 (nuts), X=1.0.

R_cap — Raise Cap

X^3.5/(1.05-X) creates an asymptote near the nuts. Also shrinks vs maniacs (keep their bluffs in).

I_mod — Auction Modifier

Lost auction = information gap. I_mod=0.5+0.5β: recovers vs maniacs whose 'strong signal' is noise.

x* — Optimal Raise

Kelly-inspired opening bet. Balances fold equity against call equity. Doesn't overbet weak edges.

Betting Decision Tree

A unified logic with no hardcoded thresholds

Facing a Bet ($C > 0$)

If $C \leq R\_cap$
→ Re-raise to x^*

If $x^* \leq C$
→ Call instead of re-raising

If $C \leq C\_cap$
→ Call (too expensive to raise)

If $C > C\_cap$
→ FOLD — cost exceeds value

We Have Initiative ($C = 0$)

If $x^* \geq min\_raise$ & can raise
→ Raise to x^*

If $x^* = 0$ or $x^* < min\_raise$
→ Check

If Cannot check
→ Fold

Auction Strategy

Honest Vickrey bidding for most hands — with gated trap for near-nut holdings

```
P_implied   =   P   +   S_eff × E³ × (1 + β/2)
(Equity-anchored implied pot)
```

```
B_true      =   P_implied × (V_gain + V_leak)
(Honest Vickrey bid)
```

```
α(E) = 0 if E ≤ 0.78
       ((E − 0.78)/0.22)²  otherwise     (Trap
activation)
```

```
B_target    =  max(0,  μ_opp − λ × σ_opp)   λ ~
U(1.5, 3.0)
```

```
B_final     =   B_true + α(E) × max(0, B_target −
B_true)
```

E^3 Scaling

Cubing equity crushes weak hands (E=0.5 → only 12.5% weight). Only strong hands get meaningful implied pots.

V_gain & V_leak

V_gain = expected |ΔEquity| from seeing one card. V_leak = what opponent learns from our remove.

Trap Gate at E > 0.78

Below 0.78 we never trap — bid honestly. Above 0.78, α^2 activates to shade bid toward B_target.

Randomised λ

λ drawn uniform [1.5, 3.0] each auction — opponent cannot reverse-engineer our threshold from bid history.

Opponent Modeling — Bayesian β Tracker

All key formulas adapt in real-time as β (bluff rate) and μ_bid evolve over 1,000 rounds

```
β  ← α × m_bluff + (1−α) × β      (EMA, α=0.05)
m_bluff = b_ratio × (1 − e_opp_proxy)
b_ratio  = opp_wager / pot
```

```
μ_bid  ← 0.15 × bid  +  0.85 × μ_bid
σ_bid  ← √(0.15 × (bid−μ_bid)²  +  0.85 × σ_bid²)
```

How β Propagates Through the Bot

- Equity: $W = E \times (1-\beta) + 1.0 \times \beta$ — higher β ↑ true win rate
- Raise cap shrinks: $\times (1 - 0.5\beta)$ — keep bluffs in the pot
- Call cap widens: $+ \beta \times \mu_{bluff} \times 1.5$ — call down bluffers
- Auction bid: $P_{implied} \times (1 + \beta/2)$ — extract more vs maniacs
- Auction modifier I_{mod} recovers from info deficit vs maniacs

Opponent Profiles

Nit (β ≈ 0.1)

Tight-passive. Few bluffs.
Our R_cap stays wide,
call cap normal.

TAG (β ≈ 0.3)

Aggressive with strong hands.
Medium adjustments.

Maniac (β ≈ 0.8)

Bluffs constantly.
Shrink raises, widen calls,
expand P_implied in auction.

Performance & Results

Leaderboard: 10th place out of qualifying teams

10th	1,000	<2%	<20ms
Final Rank	**Rounds**	**MLP MAE**	**Decision Speed**
Qualified for finals	Per game	Pre-flop accuracy	Total budget 20s

Bot Strengths

- ✓ Unified game-theoretic betting formula — no ad-hoc rules
- ✓ Equity-anchored auction bids with trap logic for monsters
- ✓ Real-time opponent bluff rate estimation (β EMA)
- ✓ Embedded MLP weights — zero I/O overhead at inference
- ✓ Board volatility adjustment prevents overconfidence
- ✓ Adaptive call cap catches maniac bluffers profitably

Speed Budget (20s total)

The Collapse — Post-Competition Analysis

After qualifying, a systematic late-round failure was identified in the bot's evaluation

Issue 1 — River Not Using Exact Equity

The MLP model was used for ALL streets including the river. On the river, the exact equity can be computed deterministically (both players' outs are fully known). The MLP's approximation error on this final street created systematic inaccuracies — the bot was making decisions on slightly wrong probability estimates at the most critical decision point.

Issue 2 — High MLP Loss on River State

The river state (2 hole cards + 4 board cards) had the highest validation loss of all 5 states. This caused the predicted equity to deviate significantly on some boards — up to 10-15% off. Since river bets are often for large amounts (shoves), these errors cascaded into unnecessary folds on strong holdings and bad calls on weak ones.

The Fix — What Should Have Been Done

On the river, enumerate all remaining cards and compute exact win probability in O(n) time. Since the river has only one unknown card distribution, this is fast and exact. The MLP should only be used for pre-flop, flop, and turn — where the search space is too large for exact methods. This trivial fix would have eliminated the erratic river folds that cost us positions.

Summary & Takeaways

Core Insight	Game-theoretic formulas beat hand-crafted heuristics — every parameter emerged from math, not rule-of-thumb thresholds.
MLP Equity	28 features + 5 state types + 200k MC samples give ~1.8% MAE on pre-flop. Fast enough for real-time play.
Auction Theory	Honest Vickrey bidding plus an E^3-gated trap activates only for near-nut holdings — prevents information leakage on medium hands.
Opponent Model	EMA-based β tracker makes all formulas adapt live — R_cap shrinks vs maniacs, call cap widens to catch their bluffs.
Lesson Learned	Use exact equity on the river. The MLP approximation introduced enough error to cause unnecessary folds on the most decisive street.